Deep Metric Learning Based on Brownian Covariance Representation for Few-Shot Hyperspectral Image Classification

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***Abstract*— Currently, few-shot learning (FSL) is widely used in image classification hyperspectral image classification (HSIC), owing to its exceptional proficiency in achieving good performance with few training samples. Although the FSL has made good progress, there are still some problems to be solved. On the one hand, existing methods rely on linear distance to learn metrics, which cannot capture the subtle similarities and differences between scarce prior samples. On the other hand, many current methods directly superimpose the features of spatial and spectral information, without deeply fusing the internal relationship between these two kinds of information. To address the aforementioned issues, a deep metric learning method based on Brownian distance covariance (DML-BDC) is proposed for few-shot HSIC. A dual-channel Brownian distance covariance feature extraction network is designed, which uses the Brownian covariance representation to model and fuse the spatial and spectral information and uses two different feature extractors to achieve the effect of information complementarity. Then, a metric loss based on Gaussian kernel distance is proposed to learn the complex nonlinear structure and subtle similarities and differences between support samples. Experiments on three benchmark datasets show that DML-BDC has advantages over the existing mainstream methods in terms of classification accuracy, generalization, and model complexity.**

***Index Terms*— Covariance representation (CR), deep metric learning (DML), few-shot, hyperspectral image classification (HSIC).**

I. Introduction

HYPERSPECTRAL images not only contain spatial information such as size, shape, and relative position of ground objects but also contain high-dimensional spectral features [1]. Moreover, the spectral resolution of hyperspectral images can reach the nanoscale, which can capture the subtle differences of different ground objects [2]. Therefore, hyperspectral imaging technology is widely used, such as lithology identification [3], medical diagnosis [4], and aerospace [5]. These applications are based on hyperspectral image classification (HSIC), which predicts the class of each pixel and generates a refined land cover classification map through pixel-by-pixel prediction [6].

Traditional machine learning methods for HSIC usually require a lot of prior information to obtain superior generalization performance [7], [8]. However, in practical applications, it is often unrealistic to obtain a large number of labeled samples, so few-shot learning (FSL) has great application value [9], [10]. FSL learns knowledge from limited prior samples and then generalizes to a large number of unlabeled samples, which can learn quickly and adapt to new tasks [11]. At present, few-shot classification algorithms based on prior knowledge can be roughly divided into three categories: cross-domain learning-based, meta-learning-based, and metric learning-based [12] methods.

Cross-domain learning plays a key role in few-shot classification tasks, which focus on transferring and applying knowledge between multiple domains with different data distributions or feature spaces [13]. This mechanism is to enhance the generalization performance of the target domain, so as to significantly reduce the dependence on labeled data in the target domain [14]. However, traditional cross-domain few-shot methods assume that the data of the base category and the new category come from the same domain or are highly similar, but this assumption is not true in practical applications. To solve this problem, Zhao et al. [15] proposed a domain adversarial prototypical network to align the source domain and the target domain while maintaining the inherent discrimination of each class. Guo et al. [16] proposed a benchmark dataset containing multiple image acquisition methods with large visual differences to more comprehensively evaluate the performance of cross-domain FSL methods. Li et al. [17] proposed a conditional adversarial strategy to align the distribution between the domains, thereby improving the generalization of the model on the target domain. Li et al. [18] constructed a cross-domain loss function based on Wasserstein distance, which effectively solved the domain shift problem. The few-shot classification performance based on cross-domain learning depends on the quality of the source domain data. If the source domain has

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noise or does not match the features and sample distribution of the target domain, then cross-domain learning will reduce the classification accuracy of the target domain.

The core idea of few-shot classification based on meta-learning is to continuously learn the general knowledge of multiple tasks, thereby improving the performance of models on new tasks [19]. Meta-learning is widely used in few-shot classification to improve machine performance in different learning tasks by learning and optimizing the learning process itself [20], [21]. Gao et al. [22] proposed a stacked convolutional model and introduced a meta-learning-based meta-optimizer to learn effective feature representations in few-shot training. To further reduce the need for labeled samples, Gao et al. [23] used principal component analysis and data augmentation methods to build different augmented features for the same sample to improve few-shot classification performance through unsupervised meta-learning methods. Hu et al. [24] constructed a 3-D ghost attention network and a double adjustment mode through meta-learning to effectively solve the problem of domain shift, and used the class covariance metric to optimize the decision boundary. However, meta-learning-based methods require a complex training process, additional computational resources, and rely on the selection of class prototypes.

Metric learning makes the potential samples of the same category more similar and the samples of different categories more different by learning a similarity matrix [25]. The few-shot algorithm of metric learning obtains the similarity between samples and effectively solves the problem of data sparsity and feature representation in a few-shot classifications [26]. To flexibly cope with different training tasks, Oreshkin et al. [27] proposed the idea of metric scaling, which dynamically adjusts the distances and weights between different samples in the metric space to improve the performance of few-shot models. To get rid of the limitation of a single-distance metric, Gao et al. [28] proposed a multidistance metric network that maps samples to several different feature spaces to learn the similarities and differences between samples more comprehensively. To obtain enough discriminative features in FSL, Li et al. [29] constructed a classification network from global to local learning, which combines metric learning and multiscale fusion to achieve good performance. Although the few-shot classification based on metric learning has made some progress, there are still limitations in extracting and modeling the spatial-spectral information and mining the distance metric of samples. The limitations are summarized as follows.

1) Hyperspectral images have irregular coverage and various categories, resulting in spatial adjacent pixels may belong to different categories. This situation has a misleading effect on models trained using spatial information.
2) Current methods do not have a unified framework for modeling spatial and spectral information to deeply fuse and mine the internal relationship between these two kinds of information, but only simply superimpose them.
3) The traditional metric loss such as contrastive loss and triplet loss is based on linear distance (Euclidean or Mahalanobis distance) to learn the metric between samples, which cannot effectively obtain and generalize the relationship between limited prior samples.

Therefore, a deep metric learning based on Brownian distance covariance representation (DML-BDC) is proposed. First, a patch-level adaptive mask module is established to mask pixels that are not similar to the central pixel and retain pixels that are highly similar to the central pixel by calculating an adaptive spectral similarity (SpeSim) threshold to alleviate the problem of inconsistent spatial coverage of ground objects. Second, a lightweight two-channel BDC extraction module is designed. The two channels use convolution and deconvolution to extract spectral features, respectively, and then calculate their BDC representations to model the spatial and spectral information into a unified framework. In addition, we also propose a Gaussian kernel metric loss, which maps the features of labeled pixels into Hilbert space to characterize their nonlinear relationship, so as to better generalize prior information to the samples to be predicted.

The contributions of our research are described below.

1) To solve the dilemma of inconsistent categories between neighboring pixels, an adaptive mask module is designed. The module can effectively identify and mask those pixels with inconsistent spectral features, which significantly improve the effective utilization of spatial information in subsequent training of the model.
2) To deeply mine and fuse spatial and spectral information, a deep feature extraction network based on Brownian covariance representation (CR) is proposed, which models these two kinds of information as a unified framework and uses two feature extraction branches to extract different information.
3) To mine the nonlinear correlation between prior samples, a Gaussian kernel metric loss is proposed to learn the distance metric of labeled samples, which can mine the subtle similarities and differences between samples.

II. Related Work

Our research mainly uses the ideas of DML and CR, so this section introduces the principles and related works of these two aspects.

A. Deeping Metric Learning

Metric learning makes the potential samples of the same category more similar and the samples of different categories more different by learning the similarity matrix [30]. DML introduces deep learning technology into metric learning, which performs nonlinear mapping on input features to learn more discriminative feature representations [31].

Deng et al. [32] designed a feature embedding algorithm for DML to effectively learn the similarity between samples. This method overcomes the problem that traditional classifiers are prone to overfitting with few-shot. Liu et al. [33] proposed a metric classifier to dynamically learn the similarity of samples, and this dynamic optimization strategy helps to better capture the relationship between samples during the training process and boost the performance of classification. Zeng et al. [34]

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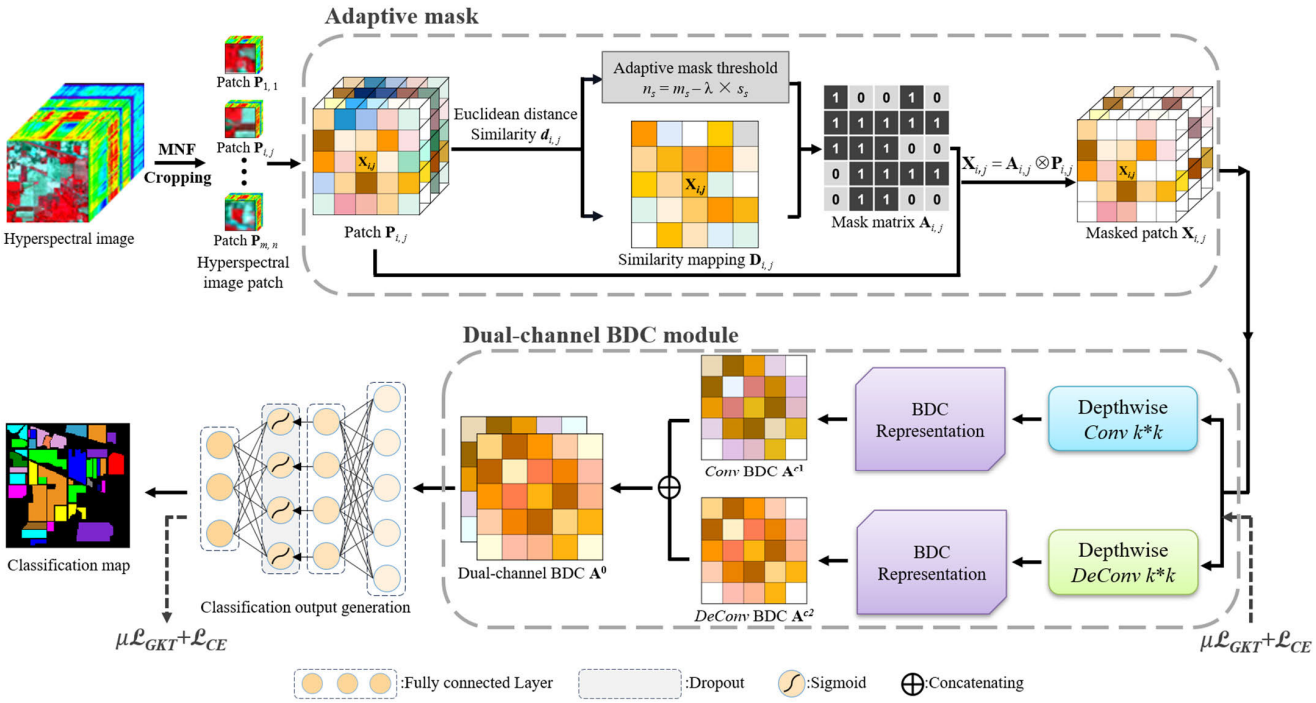


Fig. 1. Framework of DML-BDC.

designed a dual-metric relation network to learn the nonlinear relationship between image-to-category and image-to-image through a deep feature extraction network. This relational network can also learn the nonlinear relationship between samples, but this multimetric training method is prone to overfitting when the training samples are limited.

B. Distance Covariance

Distance covariance measures the degree of linear relationship between random variables [35]. Specifically, covariance represents the expectation of the overall error of two variables and reflects the degree of cooperative variation between them [36], [37]. It has been widely used in correlation analysis and classification in recent years.

Zhang et al. [38] proposed a local isomorphic CR, which extracts more discriminative features with limited information by deeply fusing spectral, spatial information, and domain construction. Torti et al. [39] represent the spatial-spectral information of each sample by calculating its local covariance matrix, which mines the spectral features and spatial neighborhood information. The traditional covariance mainly measures the linear relationship between variables, while the BDC [40] can capture the nonlinear relationship and distribution similarity between variables. Xie et al. [41] constructed a deep feature extraction network based on the Brownian CR, which learns a discriminative image representation by measuring the difference between the joint features and the edge features.

III. Proposed Methodology

Fig. 1 is the framework of DML-BDC. Given a hyperspectral image $\mathbf{X} \in \mathbb{R}^{c\times k\times g}$, where c, k, and g represent the height, width, and spectral dimensions. $\mathbf{Y}$ denotes the label corresponding to each pixel in $\mathbf{X}$. In the preprocessing part, the minimum noise fraction (MNF) [42] is used for feature extraction of $\mathbf{X}$ to reduce the spectral dimension from g to b, and then $\mathbf{X}$ is cropped into multiple patches $\mathbf{P}_{i,j} \in \mathbb{R}^{s\times s\times b}$. In the training phase, the adaptive mask module calculates the spectral Euclidean distance similarity (SpeSim) between the center pixel $\mathbf{x}_{i,j}$ and other pixels in each patch $\mathbf{P}_{i,j}$. The module sets an adaptive threshold, and other pixels whose SpeSim with the central pixel is less than the adaptive threshold are masked to alleviate the problem of inconsistent spatial coverage of ground objects. Second, a two-channel BDC representation module is designed for feature extraction and model unified spatial-spectral representation. Then, the Gaussian kernel triplet (GKT) loss maps the labeled samples into the Hilbert space to learn the nonlinear relationship between samples and generalize this relationship to unlabeled samples for more accurate classification.

A. Adaptive Mask

Image processing algorithms usually assume that spatially adjacent samples belong to the same class, so the model utilizes spatial information for training. However, the low spatial resolution of remote sensing datasets results in a higher probability that neighboring pixels belong to different classes. In this case, directly using spatial information to train the model leads to overfitting. Therefore, an adaptive mask module is designed to mask pixels that are spatially adjacent but have large differences in spectral characteristics.

As shown in Fig. 1, the patch blocks $\mathbf{P}_{i,j} \in \mathbb{R}^{s\times s\times b}$ are used as the input of the adaptive mask, and the central spectral feature $\mathbf{x}_{i,j} \in \mathbb{R}^{b}$ is selected in the patch blocks $\mathbf{P}_{i,j}$. Then, the SpeSim between $\mathbf{x}_{i,j}$ and the other pixels in $\mathbf{P}_{i,j}$ is calculated to obtain the SpeSim map matrix $\mathbf{D}_{i,j} \in \mathbb{R}^{s\times s}$, which describes the spectral distance relationship modeled by $\mathbf{x}_{i,j}$ and $\mathbf{P}_{i,j}$

$$\mathbf{D}_{i,j} = \mathrm{SpeSim}\left(\mathbf{x}_{i,j}, \mathbf{P}_{i,j}\right). \tag{1}$$

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$d_{i',j'}$ is the SpeSim value in $\mathbf{D}_{i,j}$ at (i', j'), where $i', j' = 1, \ldots, s$, $d_{i',j'}$ is as follows:

$$d_{i',j'} = \mathrm{SpeSimV}\left(\mathbf{x}_{i,j}, \mathbf{x}_{i',j'}\right) = \frac{1}{1 + \left\|\mathbf{x}_{i,j} - \mathbf{x}_{i',j'}\right\|} \tag{2}$$

where $\mathbf{x}_{i',j'}$ represents the spectral vector of samples in $\mathbf{P}_{i,j}$. $d_{i,j}$ is normalized to a uniform representation. $d_{i',j'}$ close to 1 indicates that $\mathbf{x}_{i',j'}$ is very similar to $\mathbf{x}_{i,j}$.

To adaptively select high-similarity neighborhood samples, the mask threshold n_s is set to $m_s - \gamma \times s_s$, where m_s, s_s denote the mean and standard deviation of $\mathbf{D}_{i,j}$, γ is a hyperparameter. The mask threshold n_s uses spectral information to adaptively obtain effective spatial relationships and eliminate useless spatial information.

Based on the mask threshold n_s, the adaptive mask matrix $\mathbf{A}_{i,j} \in \mathbb{R}^{s \times s}$ can be expressed as follows:

$$a_{i',j'} = \begin{cases} 1, & d_{i',j'} \geq n_s \\ 0, & d_{i',j'} \leq n_s \end{cases} \tag{3}$$

where $a_{i',j'}$ are the values in $\mathbf{A}_{i,j}$ located at (i', j'). Then, the adaptive mask matrix $\mathbf{A}_{i,j}$ is used to dot multiply $\mathbf{P}_{i,j}$ to obtain the masked patch $\mathbf{X}_{i,j} \in \mathbb{R}^{s \times s \times b}$

$$\mathbf{X}_{i,j} = \mathbf{A}_{i,j} \otimes \mathbf{P}_{i,j} \tag{4}$$

where $\otimes$ denotes dot multiplication at the pixel level.

B. Dual-Channel BDC Module

To enhance the feature representation ability, a two-channel feature extraction module based on BDC has been proposed. As shown in Fig. 1, this module takes the masked patch block $\mathbf{X}_{i,j} \in \mathbb{R}^{s \times s \times b}$ as input. To concisely represent the rich spectral features, we design a two-channel network structure to extract features, where channel 1 extracts spectral information by convolution and channel 2 extracts spectral information by deconvolution. The details are as follows:

$$\mathbf{X}^1_{i,j} = \mathrm{DepthwiseConv}_{k \times k}\left(\mathbf{X}_{i,j}\right) \tag{5}$$

$$\mathbf{X}^2_{i,j} = \mathrm{DepthwiseDeConv}_{k \times k}\left(\mathbf{X}_{i,j}\right) \tag{6}$$

where DepthwiseConv and DepthwiseDeConv denote depth convolution and depth deconvolution, respectively. Deep convolution and deconvolution are used to extract spectral information, respectively, but the original spatial size of $\mathbf{X}_{i,j}$ is not changed. $\mathbf{X}^1_{i,j}, \mathbf{X}^2_{i,j}$ are the input of the BDC representation module in channel 1 and channel 2, respectively.

To capture the correlation between different spectral bands, the similarity between different bands is learned by Euclidean distance. First, $\mathbf{X}^t_{i,j}$ is reconstructed as a 2-D matrix $\mathbf{X}^t \in \mathbb{R}^{s^2 \times \sim b}$, and then compute the Euclidean distance for spectral bands of $\mathbf{X}^t$

$$\hat{a}_{uv} = \|\mathbf{x}_u - \mathbf{x}_v\|, \quad u, v = 1, \ldots, b \tag{7}$$

where $\mathbf{x}_u$ and $\mathbf{x}_v \in \mathbb{R}^{s^2}$ denote the uth and vth spectral features of $\mathbf{X}^t$. Thus, the Euclidean distance matrix $\hat{A}^c = (\hat{a}_{uv}) \in \mathbb{R}^{b \times b}$ is further generated.

To mine the association of spatial-spectral features, a dual-channel BDC representation module is designed to model the spatial-spectral information in a unified way. It calculates the covariance of spectral information and reduces the global influence by subtracting the row and column means of the covariance (this operation utilizes spatial information), thereby highlighting local spatial-spectral information. The BDC matrix is obtained as follows:

$$\mathbf{A}^c = \hat{A}^c - \hat{A}^c_{b.} - \hat{A}^c_{.b} + \hat{A}^c_{..} \tag{8}$$

where $\hat{A}^c_{b.}, \hat{A}^c_{.b}, \hat{A}^c_{..}$ denote the row mean, column mean, and average of all elements of $\hat{A}^c$.

The corresponding BDC matrices $\mathbf{A}^{c1}$, and $\mathbf{A}^{c2}$ are obtained for channel 1 and channel 2, respectively, and the two-channel BDC representation $\mathbf{A_0}$ is obtained by stacking $\mathbf{A}^{c1}$ and $\mathbf{A}^{c2}$ in the spectral dimension.

The 3-D matrix $\mathbf{A_0}$ is flattened to a 1-D matrix. The spectral information is extracted by two fully connected layers, and the nonlinear mapping is performed by Sigmoid and Dropout, and then the prediction result is obtained by a fully connected layer. The formula is as follows:

$$\mathbf{A_1} = \mathrm{FC}_2(\mathrm{FC}_1(\mathrm{Flatten}(\mathbf{A_0}))) \tag{9}$$

$$\mathbf{A_2} = \mathrm{FC}_3(\mathrm{Dropout}(\mathrm{Sigmoid}(\mathbf{A_1}))). \tag{10}$$

Sigmoid and Dropout represent nonlinear mapping functions, and dropout prevents the model from overfitting.

C. Gaussian Kernel Metric Loss

The pixels of hyperspectral images have significant intraclass diversity and interclass similarity. It is different to utilize linear distance, such as Euclidean distance or Mahalanobis distance, to characterize the nonlinear correlation between prior pixels. In order to mine the distance metric between high-dimensional samples, we propose a Gaussian kernel metric loss based on nonlinear distance and metric learning.

Gaussian kernel metric constraint measures the similarity between samples by means of the Gaussian kernel function [43], which maps samples into high-dimensional Hilbert space to more intuitively characterize the relationship between prior samples. As shown in Fig. 2, the samples of the original feature space are mapped into the more tridimensional Hilbert feature space by Gaussian kernel function, where the interclass samples are more dispersed and the intraclass samples are more clustered. In order to learn the intraclass diversity of samples and the subtle differences between classes, Gaussian kernel metric constraint constructs the most dissimilar samples within the class as positive pairs and the most similar samples between classes as negative pairs based on triplet loss [44]. As shown in Fig. 2, triangle 2 and square 2 are constituted as negative pairs because square 2 is the closest interclass sample to triangle 2, and triangle 2 and triangle 5 are constituted as positive pairs because triangle 5 is the farthest intraclass sample from triangle 2. The formula for the GKT loss L_{GKT} is as follows:

$$L_{\mathrm{GKT}} = \frac{1}{N} \sum_{i=1}^{N} h\left(\alpha + \max_{y_p = y_i} \kappa_{ip} - \min_{y_n \neq y_i} \kappa_{in}\right) \tag{11}$$

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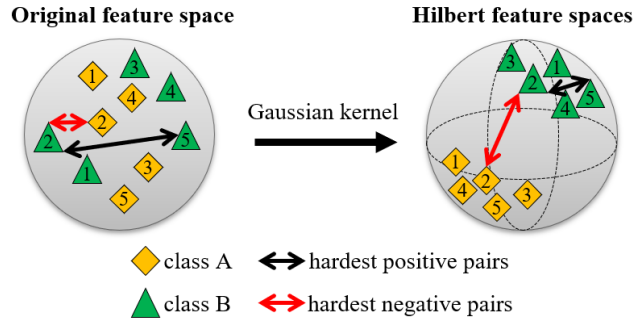


Fig. 2. Gaussian kernel metric loss.

where N denotes the number of training samples, $h(x) = \max(0, x)$ represents a hinge loss, and α represents a hyperparameter. $\max_{y_p=y_i} \kappa_{ip}$ represents the loss caused by the most dissimilar positive sample x_p to x_i, and $\min_{y_n\neq y_i} \kappa_{in}$ represents the loss caused by the most similar negative sample x_n to x_i. x_i denotes the training sample feature. x_i, x_p denotes a positive pair, and x_i, x_n denotes a negative pair. κ_{ij} represents the Gaussian kernel distance function as follows:

$$\kappa_{ij} = \sum_{k=1}^{N_k} \left(1 - e^{-\frac{\|f(x_i)-f(x_j)\|^2}{\sigma_k}}\right) \tag{12}$$

where $f(x_i)$ is the feature of sample x_i after extraction and $\|\cdot\|$ denotes the norm of the matrix. σ_k represents the standard deviation between all pairs of labeled samples.

To ensure the accuracy of the model, the cross-entropy loss L_{CE} can also improve the performance of the model as follows:

$$L_{\mathrm{CE}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c} \log p_{i,c} \tag{13}$$

where C represents the number of classes, N represents the number of training samples. $y_{i,c}$ represents the true labels, and $p_{i,c}$ represents the labels predicted by the model.

Finally, the weighting of the L_{GKT} and the L_{CE} is used as the final loss function, as follows:

$$L = \mu L_{\mathrm{GKT}} + L_{\mathrm{CE}} \tag{14}$$

where μ represents a hyperparameter used to trade off L_{GKT} and L_{CE}.

IV. Experiments

A. Experimental Data

We conduct experiments on Indian, Houston 2013, and Salinas to verify the effectiveness of DML-BDC. Fig. 3 shows the pseudo-color maps, ground-truth maps. Table I shows the information of these three datasets. Table II shows the color bars, the number of training samples, and test samples for each category of the Indian Pines, Houston 2013, and Salinas.

B. Experimental Settings

1) Parameter Setting: The number of training iterations is 65 for all datasets. Except for the generalization experiment, the other experiments randomly select five samples per class as the training set. We use several approaches to test the performance of DML-BDC: generalization experiments, hyperparameter experiments (including dimensionality reduction ratio, and mask threshold), feature extraction experiments, and computational complexity analysis.

TABLE I
Parameters of the Dataset

Data set	Indian Pines	Houston 2013	Salinas
Sensor	AVIRIS	ITRES CASI 1500	AVIRIS
Image size	145145	3491905	512217
Number of bands	200	144	204
Wavelength range (nm)	400-2500	364-1046	400-2500
Spatial resolution (m)	20	2.5	3.7
Number of classes	16	15	16
Number of samples	10249	15029	54129

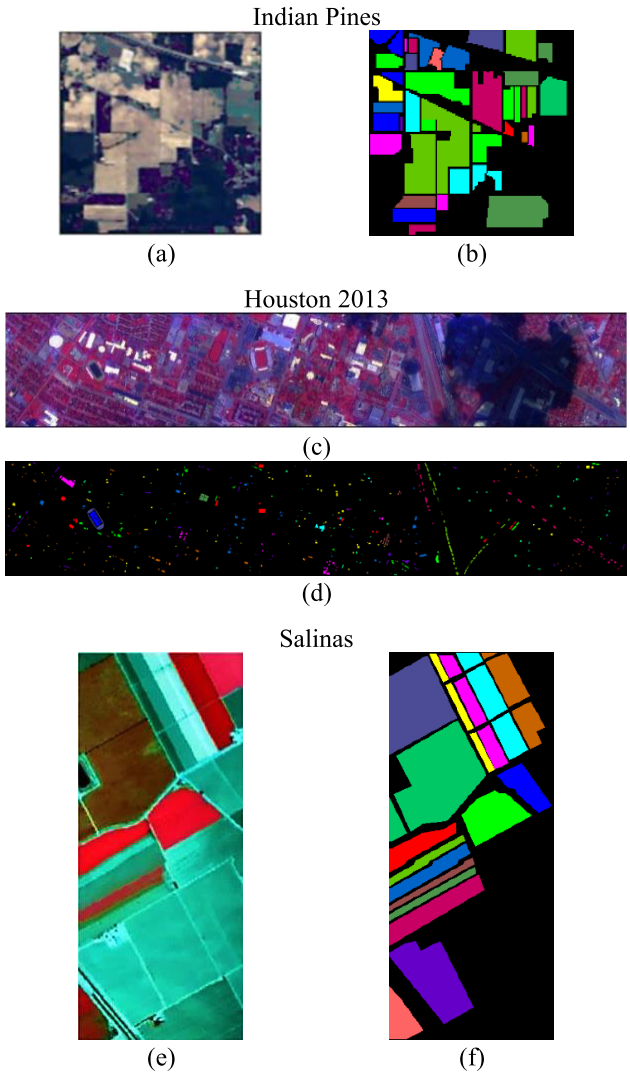


Fig. 3. (a), (c), and (e) False-color maps. (b), (d), and (f) Ground-truth maps.

2) Performance Comparison: To compare the performance of the algorithm, seven algorithms are selected to compare with DML-BDC, including two few-shot metric learning methods: similarity-based deep metric module (SDMM) [32] and few-shot learning with class-covariance metric (CMFSL) [37]; a DML method: composite neighbor-aware convolutional metric network (CNCMN) [33]; four few-shot classification methods: deep cross-domain few-shot learning (DCFSL) [17], spectral–spatial Siamese network (S3Net) [9], multistage relation network with dual-metric (DM-MRN) [34], and self-pooling transformer (SPFormer) [10].

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TABLE II
NUMBER OF TRAINING AND TEST SAMPLES FOR EACH CATEGORY OF THE THREE DATASETS

	Indian Pines			Houston 2013			Salinas		
	Class	Train.	Test.	Class	Train.	Test.	Class	Train.	Test.
1	Alfalfa	5	41	Healthy grass	5	1246	Brocoli green weeds 1	5	2004
2	Corn-notill	5	1423	Stressed grass	5	1249	Brocoli green weeds 2	5	3721
3	Corn-mintill	5	825	Synthetic grass	5	692	Fallow	5	1971
4	Corn	5	232	Trees	5	1239	Fallow rough plow	5	1389
5	Grass-pasture	5	478	Soil	5	1237	Fallow smooth	5	2673
6	Grass-trees	5	725	Water	5	320	Stubble	5	3954
7	Grass-pasture-mowed	5	23	Residential	5	1263	Celery	5	3574
8	Hay-windrowed	5	473	Commercial	5	1239	Grapes untrained	5	11266
9	Oats	5	15	Road	5	1247	Soil vinyard develop	5	6198
10	Soybean-notill	5	967	Highway	5	1222	Corn senesced green weeds	5	3273
11	Soybean-mintill	5	2450	Railway	5	1230	Lettuce romaine 4wk	5	1063
12	Soybean-clean	5	588	Parking Lot 1	5	1228	Lettuce romaine 5wk	5	1922
13	Wheat	5	200	Parking Lot 2	5	464	Lettuce romaine 6wk	5	911
14	Woods	5	1260	Tennis Court	5	423	Lettuce romaine 7wk	5	1065
15	Buildings-Grass-Trees	5	381	Running Track	5	655	Vinyard untrained	5	7263
16	Stone-Steel-Towers	5	88				Vinyard vertical trellis	5	1802

TABLE III
CLASSIFICATION ACCURACY OF INDIAN PINES

Class	DCFSL	CMFSL	SDMM	CNCMN	S3Net	DM-MRN	SPFormer	DML-BDC
1	**100.00**	**100.00**	54.05	**100.00**	**100.00**	**100.00**	**100.00**	97.62
2	26.56	31.06	43.28	56.92	38.37	54.20	71.61	**96.44**
3	56.61	53.09	40.40	71.15	43.03	39.64	53.58	**94.81**
4	94.83	94.40	27.79	99.13	96.98	77.64	**99.57**	70.90
5	82.64	79.50	74.80	81.79	83.05	47.20	82.22	**100.00**
6	95.17	95.03	89.83	**99.86**	98.76	82.88	94.21	94.64
7	**100.00**	**100.00**	21.98	**100.00**	**100.00**	**100.00**	**100.00**	95.83
8	94.50	97.67	97.54	91.75	**100.00**	99.79	**100.00**	96.14
9	**100.00**	**100.00**	13.16	86.66	**100.00**	**100.00**	**100.00**	68.18
10	65.77	62.87	36.76	58.63	59.15	66.46	**71.15**	64.45
11	66.00	61.51	55.04	67.59	77.31	68.11	62.49	**85.41**
12	44.56	32.31	23.89	**66.67**	60.37	31.70	62.93	59.23
13	99.00	99.50	76.26	**100.00**	**100.00**	**100.00**	98.00	97.56
14	89.13	85.87	91.40	90.15	92.54	79.76	**98.02**	94.99
15	47.77	59.84	61.07	65.87	80.84	94.04	78.22	**94.40**
16	**100.00**	**100.00**	82.86	**100.00**	96.59	95.70	**100.00**	81.48
OA (%)	66.69±2.94	65.01±3.82	54.73±5.64	74.17±0.89	72.49±3.55	66.96±3.74	75.98±3.39	**85.28±3.12**
AA (%)	78.91±1.11	78.29±1.52	55.63±2.48	83.51±1.52	82.94±1.53	77.32±1.96	85.75±1.90	**87.01±1.76**
$\kappa \times 100$	62.51±3.01	60.51±3.96	49.83±6.08	70.97±1.26	68.75±3.88	62.86±3.82	72.86±3.74	**83.29±2.92**

3) Evaluation Metrics and Operating Platforms: The metrics to evaluate the classification performance are overall accuracy (OA), average accuracy (AA), and kappa (κ) coefficient. All experiments shown below are implemented on Windows 11 using Pytorch 1.13.0 and run on the platform with Intel[1] Core i7-12700f CPU and NVIDIA 3090 GPU.

C. Classification Results

The performance of DML-BDC is verified by comparing the classification result of DML-BDC with the other seven algorithms. Each experiment was performed five times, and the median and standard deviation of their OA, AA, and κ were taken. Quantitative analysis was carried out by OA, AA, κ coefficient, and classification accuracy of each category, and qualitative analysis was carried out by classification map.

1) Indian Pines: The classification accuracy of Indian Pines is shown in Table III, where DML-BDC achieved the highest OA, AA, and κ compared to the other algorithms. The OA value is 9.30%–30.55% higher than the other seven algorithms, the AA value is 1.26%–31.38% higher than other methods, and the κ value is 10.43%–33.46% higher than other methods. Corn-notill and Soybean-mintill are the two categories with the largest number of samples. In FSL, the features learned by the model from the categories with a large number of samples are often not universal and cannot be generalized to other similar samples, so the classification accuracy of few-shot algorithms for these two categories is low. However, the accuracy of DMLBDC in classifying Corn-notill and

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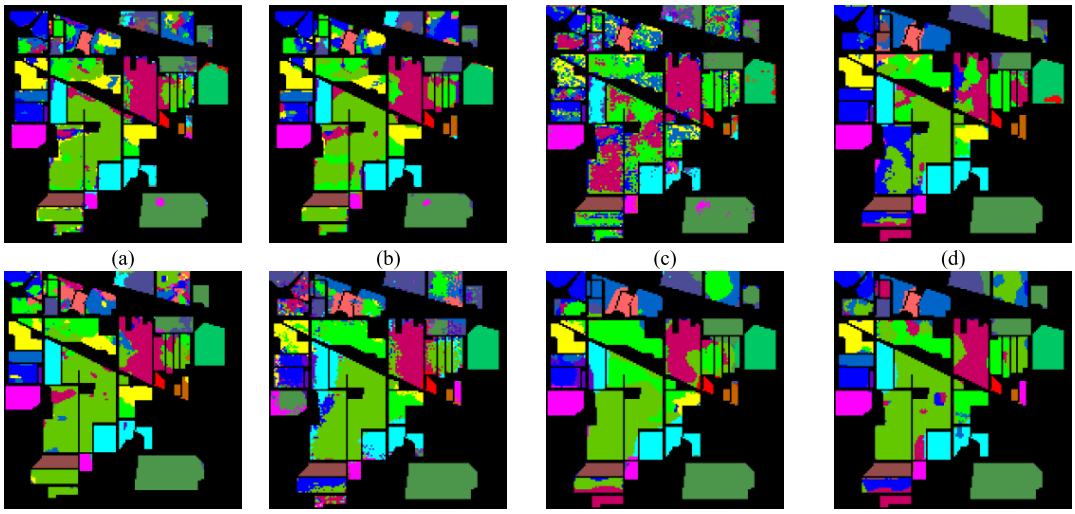


Fig. 4. Classification map of Indian Pines. (a) DCFSL (66.69%). (b) CMFSL (65.01%). (c) SDMM (54.73%). (d) CNCMN (74.17%). (e) S3Net (72.49%). (f) DM-MRN (66.96%). (g) SPFormer (75.98%). (h) DML-BDC (85.28%).

TABLE IV
CLASSIFICATION ACCURACY OF HOUSTON 2013

Class	DCFSL	CMFSL	SDMM	CNCMN	S3Net	DM-MRN	SPFormer	DML-BDC
1	**98.96**	98.88	87.52	72.81	91.49	94.59	90.69	77.47
2	80.62	81.91	88.35	56.24	94.64	93.71	71.18	**95.56**
3	92.63	97.69	99.71	99.71	**100.00**	99.14	99.13	98.72
4	88.30	91.12	97.02	74.84	92.74	94.44	92.90	**99.82**
5	98.22	97.01	82.80	98.71	89.49	86.87	**100.00**	95.37
6	79.69	80.94	**100.00**	87.85	**100.00**	84.62	76.25	96.89
7	71.97	84.48	69.69	56.96	85.43	64.01	86.30	**91.75**
8	39.95	60.29	53.43	41.94	33.49	82.78	45.12	**82.99**
9	82.20	**83.48**	81.88	54.81	73.06	64.46	61.59	69.72
10	51.47	53.36	69.51	86.84	39.28	75.76	**95.25**	66.65
11	69.11	65.77	52.69	79.45	**83.50**	53.56	75.28	76.71
12	**86.32**	50.08	78.40	59.40	35.42	74.81	55.13	77.69
13	85.99	91.81	48.22	69.25	**93.53**	57.74	85.78	43.47
14	84.87	87.94	80.88	**100.00**	**100.00**	81.98	**100.00**	80.11
15	**100.00**	96.95	97.04	**100.00**	**100.00**	99.09	**100.00**	**100.00**
OA (%)	79.08±2.67	79.46±1.43	76.47±3.08	72.35±1.78	76.56±2.72	78.61±3.98	80.23±2.13	**81.80±2.01**
AA (%)	80.69±2.41	81.45±1.21	79.14±1.31	75.92±0.63	80.80±2.22	80.50±1.56	82.31±1.99	**83.53±1.21**
$\kappa \times 100$	77.38±2.88	77.79±1.52	74.56±2.71	70.11±1.98	74.69±2.93	76.91±3.68	78.65±2.30	**80.36±1.92**

Soybean-mintill is much higher than that of the comparison algorithm. This is because the Gaussian kernel metric loss adopted by DMLBDC can learn more representative class features and avoid overfitting. Fig. 4 shows the classification map. From a macro perspective, DML-BDC has the least misclassified pixels. From the microscopic point of view, other algorithms misclassify the Soybean-mintill area in the middle of the image in large numbers and irregularly, but DML-BDC has fewer misclassifications and is more clustered.

2) Houston 2013: The classification accuracy of Houston 2013 is shown in Table IV. Compared with the other seven algorithms, DML-BDC achieved the highest OA, AA, and κ values. The OA, AA, and κ of DML-BDC are 1.57%–9.45%, 1.22%–7.61%, and 1.71%–10.25% higher than those of other algorithms, respectively. Fig. 5 shows the prediction map. In Houton 2013, the sparse distribution of labeled pixels leads to the phenomenon that the whole area is misclassified. The enlarged area in Fig. 5 is the tennis court category, but SDMM, CNCMN, and S3Net misclassified the entire area as soil, and SPFormer misclassified the entire area as residential. These four algorithms all learn the neighborhood structure between samples through spatial information, so that all samples in adjacent regions are misclassified. The adaptive mask proposed by DML-BDC is able to avoid this phenomenon. DML-BDC can mask the spatially adjacent samples with different spectral features to avoid misclassification caused by inconsistent spatial distribution of ground objects. DCFSL, CMFSL, and DM-MRN appear local misclassification, and DML-BDC does not misclassify.

3) Salinas: The classification accuracy of Salinas is shown in Table V, where DML-BDC achieved the highest OA, AA, and κ values compared to the other seven algorithms. The

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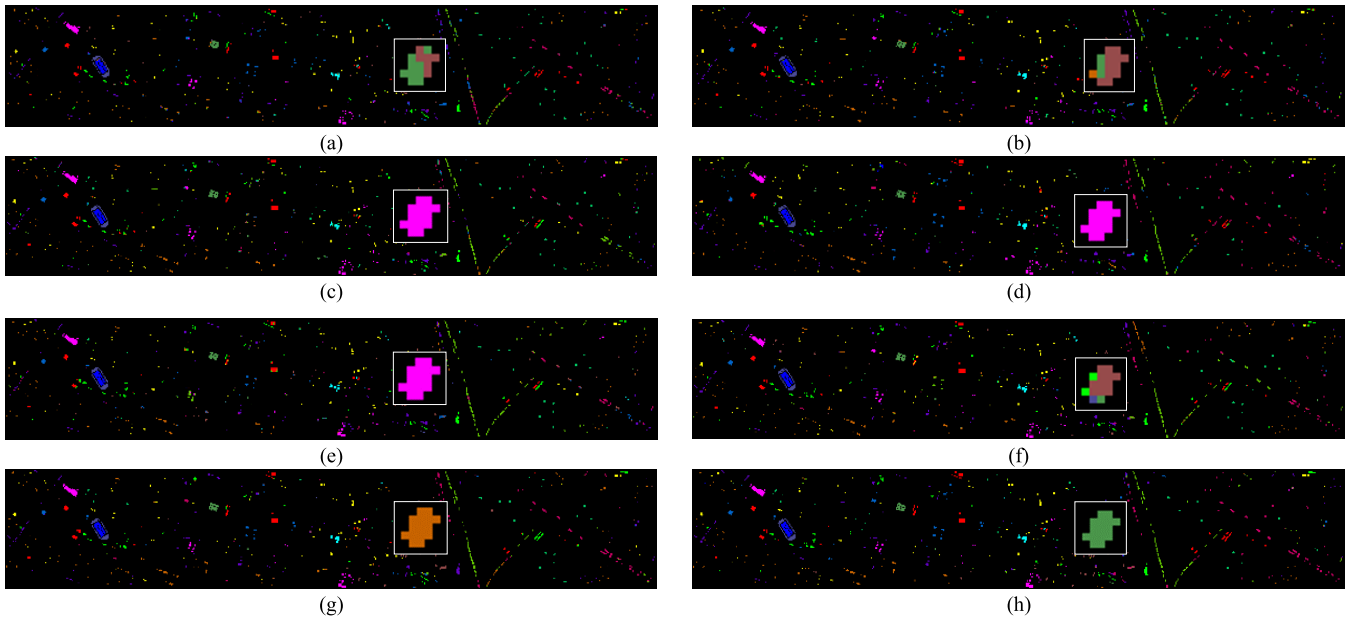


Fig. 5. Classification map of Houston 2013. (a) DCFSL (79.08%). (b) CMFSL (79.46%). (c) SDMM (76.47%). (d) CNCMN (72.35%). (e) S3Net (76.56%). (f) DM-MRN (78.61%). (g) SPFormer (80.23%). (h) DML-BDC (81.80%).

TABLE V
CLASSIFICATION ACCURACY OF SALINAS

Class	DCFSL	CMFSL	SDMM	CNCMN	S3Net	DM-MRN	SPFormer	DML-BDC
1	**100.00**	99.00	99.55	93.77	98.60	**100.00**	**100.00**	**100.00**
2	99.19	99.95	99.62	**100.00**	96.88	**100.00**	**100.00**	99.84
3	96.75	99.44	95.12	**100.00**	99.85	99.85	84.83	**100.00**
4	99.78	99.42	94.29	94.10	**99.93**	97.20	99.78	95.66
5	94.09	92.93	99.77	98.39	96.30	**100.00**	99.25	98.34
6	**100.00**	**100.00**	**100.00**	99.47	**100.00**	99.85	99.22	**100.00**
7	99.92	99.61	97.64	99.72	99.94	**100.00**	99.92	99.86
8	69.53	68.92	77.02	82.31	**94.43**	92.03	77.57	84.53
9	99.94	**100.00**	99.07	**100.00**	99.90	99.58	**100.00**	99.34
10	93.37	93.89	93.15	97.77	93.77	**100.00**	93.37	97.68
11	99.44	99.81	74.56	99.44	98.31	89.02	**100.00**	**100.00**
12	99.95	99.53	99.33	99.74	87.67	**100.00**	95.79	98.77
13	98.90	98.46	84.51	98.90	97.80	99.56	**100.00**	**100.00**
14	98.87	98.59	98.97	95.31	96.90	97.16	**100.00**	97.68
15	75.60	67.73	65.04	**86.62**	62.26	54.35	81.50	83.98
16	99.39	99.00	98.11	99.45	98.61	89.88	**100.00**	**100.00**
OA (%)	89.41±1.68	88.26±1.86	88.35±0.95	93.71±0.34	92.30±1.19	87.80±0.85	91.63±1.18	**94.13±1.63**
AA (%)	95.30±0.76	94.77±0.59	92.23±0.66	96.56±0.21	95.07±0.91	94.31±0.53	95.70±0.75	**97.23±1.41**
$\kappa \times 100$	88.26±1.84	86.96±2.06	87.04±1.06	93.01±0.38	91.40±1.33	86.49±0.95	90.70±1.31	**93.46±1.80**

OA value is 0.42%–6.33% higher than other algorithms, the AA value is 0.67%–5% higher than other algorithms, and the κ value is 0.45%–6.97% higher than other algorithms. Fig. 6 shows the prediction map. From a macro perspective, DML-BDC has the least misclassified pixels and is smoother. DCFSL, CMFSL, SDMM, S3Net, DM-MRN, and SPFormer all misclassify samples from the upper-right area of corn senesced green weeds as romaine lettuce 4 weeks because the spectral features are particularly similar between these two classes. DML-BDC does not misclassify corn senesced green weeds as romaine lettuce 4 weeks, because the GKT loss in DML-BDC proposes that the idea of learning features by combining the most dissimilar intraclass samples into positive pairs and the most similar interclass samples into negative pairs can learn more subtle feature differences between samples.

D. Hyperparameter Analysis

1) For the Scale of the Dimensionality Reduction: Before training, MNF reduces the dimension of the features to remove the noise in the spectral information. In this experiment, different dimensionality reduction ratios are set to train the model, and then OA, AA, and κ are compared. A total of five experiments are included, ranging from 0.1 to 0.5. Table VI shows that the Indian Pines obtained the highest OA, AA, and κ at 0.4. Houston 2013 and Salinas achieve the best OA, AA,

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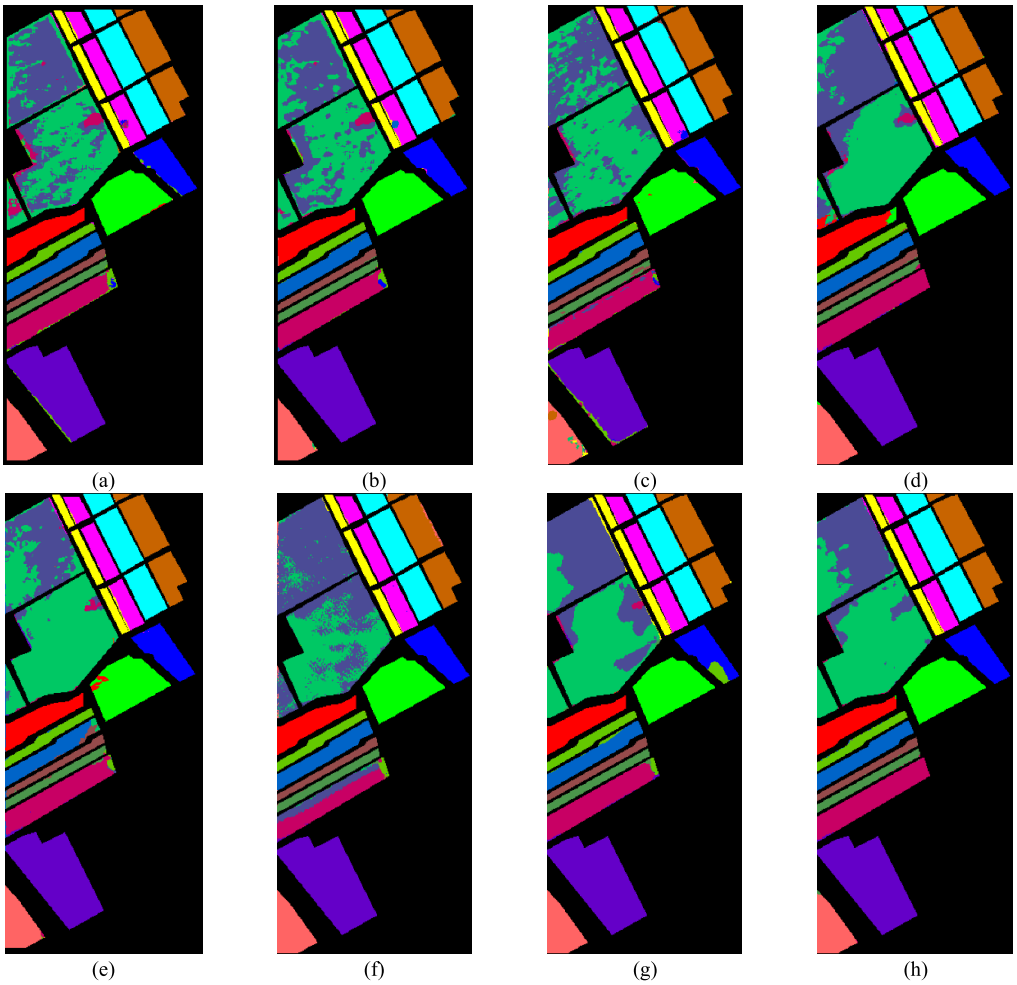


Fig. 6. Classification map of Salinas. (a) DCFSL (89.41%). (b) CMFSL (88.26%). (c) SDMM (88.35%). (d) CNCMN (93.71%). (e) S3Net (92.30%). (f) DM-MRN (87.80%). (g) SPFormer (91.63%). (h) DML-BDC (94.13%).

and κ values at 0.3 and 0.4, respectively, so their ratios are set to 0.3 and 0.4, respectively.

2) For the Masking Threshold: In the adaptive sampling module, the masking threshold is set as $n_s = m_s - \gamma \times s_s$. The selection of γ determines the masking effect of the patch. Therefore, we set up seven experiments to select the best γ value, where γ ranges from 0.1 to 0.4. Fig. 7(a) shows that the three datasets achieve the highest OA value at 0.25.

3) For the Patch Size: The patch size has an important impact on the experimental accuracy, so it needs to be weighed according to the characteristics of the experimental data. Fig. 7(b) shows OA with different patch sizes on the three datasets. On the three datasets, the OA value of patch size = 9 is the best. When patch size is less than 9, OA is always growing. When the patch size is larger than 9, the OA value is the same as the accuracy of patch size = 9 or there is a slow decrease.

4) For the Hyperparameter μ Used to Trade-Off L_{GKT} and L_{CE}: The loss function of DML-BDC is composed of Gaussian kernel metric loss and cross-entropy loss, and μ is a hyperparameter to trade off these two losses. Fig. 7(c) shows the OA with different μ values. According to Fig. 7, $\mu = 0.4$ is set for Indian, $\mu = 0.6$ for Houston 2013, and $\mu = 0.3$ for Salinas.

E. Ablation Experiments

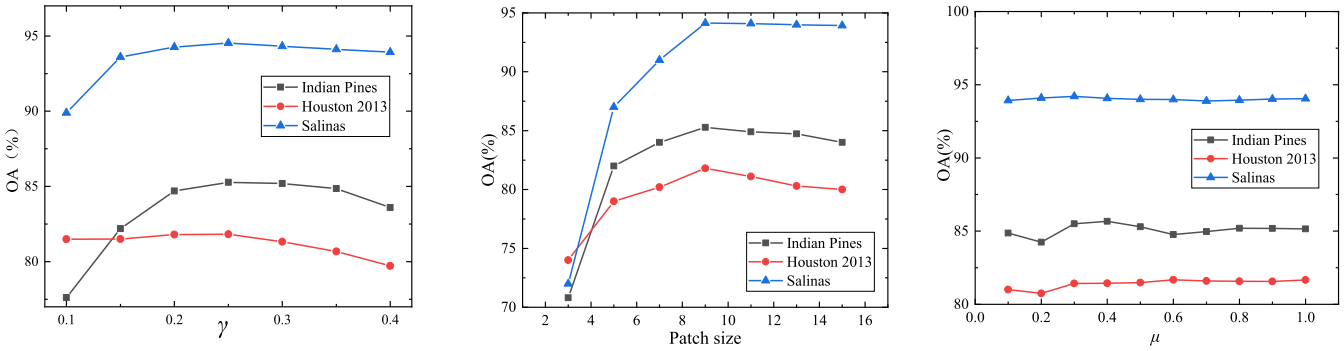
1) Effectiveness of Adaptive Mask: In DML-BDC, the adaptive mask part alleviates the problem of spatial coverage inconsistency of ground objects by masking pixels that are not similar to the center pixel. To verify the effectiveness of the adaptive mask module, we remove the adaptive mask module of the DML-BDC method for experiments. Table VII shows that the adaptive mask module improves the performance of DML-BDC on the three hyperspectral images. The adaptive mask module improves κ by 3.51%, 1.94%, and 0.32% for Indian, Houston 2013, and Salinas.

2) Effectiveness of BDC Representation: In DML-BDC, the BDC representation captures more representational sample features. To verify the effectiveness of the BDC representation, we change the BDC to the classical CR. Table VII shows that BDC representation makes DML-BDC significantly improve the accuracy of the three datasets. The κ of the Indian Pines dataset is increased by 11.67%, the κ of the Houston 2013 dataset is increased by 5.53%, and the κ of the Salinas dataset is increased by 3.26%. BDC representation makes

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TABLE VI
CLASSIFICATION ACCURACY OF DIFFERENT DIMENSIONALITY REDUCTION RATIOS

	Metrics \ Scale	0.1	0.2	0.3	0.4	0.5
Indian Pines	OA (%)	72.76	76.05	84.26	**85.28**	86.12
	AA (%)	79.51	81.32	87.46	**87.01**	84.11
	κ ×100	69.33	73.14	82.05	**83.29**	84.15
Houston 2013	OA (%)	75.36	80.24	**81.80**	81.26	81.59
	AA (%)	78.6	83.25	**83.53**	82.75	82.57
	κ ×100	73.44	78.67	**80.36**	79.77	80.14
Salinas	OA (%)	91.65	93.75	94.10	**94.13**	93.35
	AA (%)	95.75	96.68	96.94	**97.23**	96.42
	κ ×100	90.7	93.03	93.42	**93.46**	92.60



(a) (b) (c)

Fig. 7. Parameter analysis experiment. (a) OA with different γ values (Adaptive mask module). (b) OA obtained with different patch sizes. (c) OA with different μ.

TABLE VII
RESULTS OF ABLATION EXPERIMENTS

	Adaptive mask	BDC	L	OA (%)	AA (%)	κ ×100
Indian Pines	×	√	√	82.09	78.56	79.78
	√	CR	√	74.97	81.17	71.62
	√	√	L_{CE}	82.86	85.31	80.64
	√	√	L_{CE}+L_T	83.62	86.99	81.44
	√	√	√	**85.28**	**87.01**	**83.29**
Houston 2013	×	√	√	80.01	81.86	78.42
	√	CR	√	76.66	79.93	74.83
	√	√	L_{CE}	81.45	83.01	79.98
	√	√	L_{CE}+L_T	81.74	83.54	80.29
	√	√	√	**81.80**	**83.53**	**80.36**
Salinas	×	√	√	93.84	96.55	93.14
	√	CR	√	91.21	95.65	90.20
	√	√	L_{CE}	94.06	96.50	93.10
	√	√	L_{CE}+L_T	94.10	97.05	93.34
	√	√	√	**94.13**	**97.23**	**93.46**

DML-BDC have a significant improvement in the accuracy of the three datasets, indicating that BDC representation can well mine the nonlinear correlation between spectral bands.

3) Effectiveness of GKT Loss: In DML-BDC, GKT loss maps prior sample features to Hilbert space to characterize nonlinear relationships between samples. The loss function in DML-BDC is the weighting of the L_{GKT} and the L_{CE}. To verify the effectiveness of the L_{GKT}, we set up two experiments: the retention-only L_{CE}, the weighting of the L_{CE} and the triplet loss L_T. The experimental results in Table VII show that the experiment with only L_{CE} has the lowest accuracy among the three loss experiments, the second lowest is the weighting of the L_T and the L_{CE}, and the loss L of DML-BDC (the weighting of L_{GKT} and L_{CE}) has the highest accuracy. It can be seen that using Gaussian kernel distance as the metric between samples can improve the classification accuracy because it can obtain more accurate nonlinear features.

F. Generalization Performance

To verify the generalization of DML-BDC, we set 1–5 (with a step size of 1) and 5–25 (with a step size of 5) labeled samples on each category for training. The experimental results are shown in Fig. 8, which shows the OA obtained by training with different numbers of labeled samples on the three datasets. Five experiments are set up in the range of 5–25 (with a step size of 5) training samples. For Fig. 8(a) Indian Pines, the OA of DML-BDC is 1.79%–8.73% higher than

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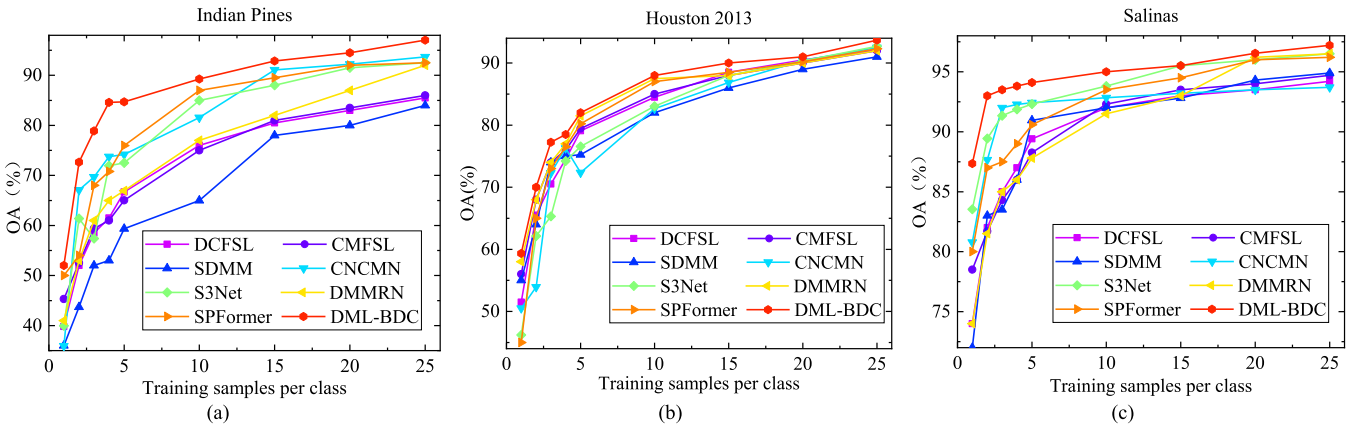


Fig. 8. OA for different training sample sizes. For (a) Indian Pines, (b) Houston 2013, (c) Salinas.

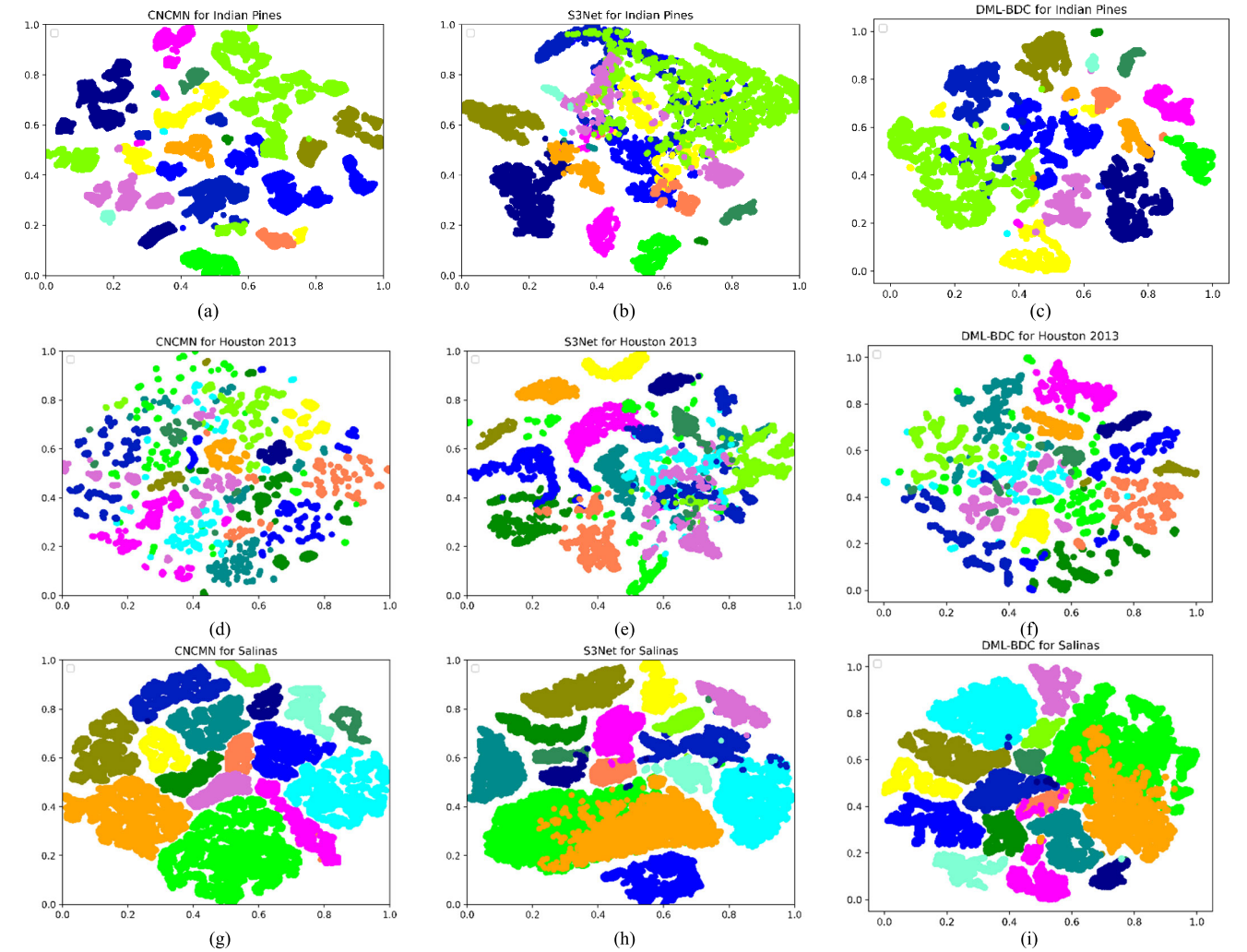


Fig. 9. Scatter plots of three algorithms in three datasets. (a) CNCMN, (b) S3Net, and (c) DML-BDC for Indian Pines. (d) CNCMN, (e) S3Net, and (f) DML-BDC for Houston 2013. (g) CNCMN, (h) S3Net, and (i) DML-BDC for Salinas.

that of the second place in five experiments. For the Houston 2013 dataset in Fig. 8(b), DML-BDC has the highest classification accuracy among the five experiments. For Fig. 8(c) Salinas, DML-BDC leads the classification accuracy in all five experiments. In order to further verify the FSL performance of the algorithm, the experimental accuracy with fewer training samples is compared. Fig. 8(a)–(c) shows the OA trained with 1–5 labeled samples (with step size of 1) for the three datasets, respectively. DML-BDC achieves the highest accuracy on all three datasets, especially on Indian Pines. The Gaussian kernel metric loss proposed by DML-BDC is robust due to its ability to extract subtle differences between samples.

G. Feature Separability Analysis

To verify the feature extraction ability of DML-BDC, we use t-SNE to map the features into a 2-D space to compare

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TABLE VIII
COMPARISON OF OPERATIONAL EFFICIENCY AND COMPUTATIONAL COMPLEXITY

		DCFSL	CMFSL	SDMM	CNCMN	S3Net	DMMRN	SPFormer	DML-BDC
Indian Pines	Training time (s)	1507.01	1740.28	217.40	**7.72**	9.32	123.36	11.91	10.26
	Testing time (s)	1.56	0.98	36.26	2.56	5.15	14.22	3.22	**0.0027**
	Params	4268994	612179	38975	**13824**	164464	181563	86296	3757765
Houston 2013	Training time (s)	1343.90	1388.16	11.34	13.96	15.33	275.24	654.77	**10.37**
	Testing time (s)	1.52	1.33	65.77	3.83	8.98	410.52	3.54	**0.0026**
	Params	4264360	610027	37005	**13312**	117851	181782	46735	2052868
Salinas	Training time (s)	1875.02	1939.69	994.57	**7.80**	27.95	143.32	38.47	13.13
	Testing time (s)	7.35	6.11	193.71	19.46	28.28	68.75	12.38	**0.0045**
	Params	4269394	614006	40385	**15217**	164464	183062	86456	4007725

the feature separability of the algorithms. CNCMN and S3Net are selected to compare with DML-BDC in the experiment, and the feature scatter plots of each algorithm on the three datasets are compared. Fig. 9(a)–(c) shows the feature scatter plots of the three algorithms for Indian pines, which shows that CNCMN has the best feature extraction effect, DML-BDC also achieves a good extraction effect, and S3Net has the worst feature extraction ability. Fig. 9(d)–(f) shows the feature scatter plots of the three algorithms for Houston 2013. It can be seen that the feature extraction effect of CNCMN is poor, S3Net has the effect of clustering but is very confusing, and DML-BDC has a certain effect of clustering but is still not dense enough. Fig. 9(g)–(i) shows the feature scatter plots of the three algorithms for Salinas, respectively. It can be seen that CNCMN, S3Net, and DML-BDC have achieved a good feature extraction effect. DML-BDC has advantages in feature extraction ability on Indian pines and Salinas but has a poor feature extraction effect on Houston 2013. This indicates that DML-BDC has advantages in processing densely distributed data such as Indian Pines and Salinas, but its effect is not prominent in processing sparsely distributed data such as Houston 2013.

H. Complexity Analysis

We compare the computational complexity of DML-BDC with seven algorithms using the following metrics: training time, testing time, and parameters. Table VIII shows that CNCMN has the smallest number of parameters, DCFSL has the largest number of parameters, and DML-BDC has the second-largest number of parameters after DCFSL. The dual-channel BDC module of DML-BDC uses two different branches to process features and then concatenates the features of the two channels so that the sum of the two branches results in an increased number of parameters. However, this operation can be more targeted to learn different information and avoid processing complex and diverse features on a single channel, thus improving computational efficiency. Although the number of parameters is large, the training and testing time of DML-BDC is shorter than most of the algorithms. This is because the adaptive mask module masks some samples to effectively filter out redundant information and reduce unnecessary calculations. In addition, the GKT loss is able to optimize the feature space in a more targeted way, thus improving the convergence speed of the loss function.

V. CONCLUSION

This article proposes a deep metric method based on dual-channel BDC representation for few-shot HSIC. First, an adaptive masking module is proposed for masking pixels with large spectral differences to alleviate the problem of inconsistent local ground object distribution. Second, a two-branch BDC feature extraction network is designed for concisely representing rich spectral information. Finally, a Gaussian kernel metric constraint is proposed to capture subtle differences in samples between classes and insignificant similarities in samples within classes. Various experiments are set up to verify the superior performance of DML-BDC. Although DML-BDC has a good advantage over the current methods, its number of parameters is large. We will further optimize the feature extraction module of DML-BDC to reduce the parameters of the model.

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